

# ME2 — ALFRED: Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

3 October 2026 · Edward Vincent Duero · Tiny Conformer · Hugging Face · 1× A100 · Raspberry Pi 5

## Model

|                                                                        |                |
|------------------------------------------------------------------------|----------------|
| Mic · 16 kHz 40-bin log-Mel                                            |                |
| Tiny Conformer · 4 × 96-dim blocks<br>full-context · attention pooling |                |
| Intent head                                                            | Slot CTC head  |
| In-scope gate                                                          | Raspberry Pi 5 |

| Item                 | Value                  |
|----------------------|------------------------|
| Parameters / weights | 0.936 M · 4.55 MB ONNX |

## Validation on the Raspberry Pi 5

| Item                 | Value                             |
|----------------------|-----------------------------------|
| Keyword / intent acc | 90.1 % / 52.5 %                   |
| False-accept rate    | 6.2 % (1 / 16 OOS)                |
| Latency p95 / RTF    | 12.4 ms / 0.0063                  |
| Runtime              | onnxruntime · 2 thr · 543 MB peak |

## Dataset

| Item         | Value                                                            |
|--------------|------------------------------------------------------------------|
| Source       | HF benchmark @ da92a79...ed006<br>+ supplement @ c376ada...d25b5 |
| Hours / utts | 23.73 h / 35,810 prepared                                        |
| Speakers     | 622 documented                                                   |
| Labels       | 19 intents · 6 slots · OOS                                       |

## Training on the A100 cluster

| Item         | Value                                |
|--------------|--------------------------------------|
| Cluster      | 1 × NVIDIA A100 · CUDA:0             |
| Objective    | CE + 0.5 CTC + 0.5 BCE               |
| Optimiser    | AdamW · TPE HPO (6/model)            |
| Steps / loss | 35 epochs · batch 64 · seeds 231/947 |

# To Be Submitted

|                          |                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>GitHub repository</b> | github.com/vincedevs/AI231-ME · public · project: ME2/ · MIT code                                                                    |
| <b>Dataset location</b>  | HF airimonda/ai231-me2-voice-commands @ da92a79...ed006<br>HF vincedevs/alfred-vcm-supplement @ c376ada...d25b5 · public             |
| <b>A100 cluster</b>      | 1 × NVIDIA A100 · CUDA:0 · 2.44 h · seeds 231 and 947                                                                                |
| <b>Model weights</b>     | Tiny Conformer ONNX · 4.55 MB · ME2/alfred/assets/models/vcm/<br>artifact 21c7b228fd63-20559251ca67 · checksummed · terms documented |

## What the reviewer will look for

| # | Item                                         | Status  |
|---|----------------------------------------------|---------|
| 1 | Repo public, one-command reproduction        | PARTIAL |
| 2 | Dataset licensed and citable (DOI)           | PARTIAL |
| 3 | Training logs + final checkpoint committed   | PARTIAL |
| 4 | Pi 5 latency reproduced by the posted script | DONE    |
| 5 | Held-out test set, unseen speakers           | DONE    |
| 6 | Baseline of comparable size compared         | PARTIAL |